

LIFEWS AgriShine(TM) GrowMeal(TM)

Primary 4-6 ASSESSMENT & QUIZ SYSTEM

STUDENT ASSESSMENT BOOKLET

This booklet assesses four dimensions: knowledge, practical competence, investigation/data skills, and mission-based problem solving.

Knowledge	20%
Practical competence	25%
Investigation/Data	30%
Mission problem solving	25%

Name: _____ Class: _____

School: _____ Term/Date: _____

ASSESSMENT 01 - Raised Food Garden

SECTION A - KNOWLEDGE

AQ-201 Define or explain one core idea related to raised-bed crop growth and spacing.

AQ-202 Why is a fair comparison important when studying the raised bed?

AQ-203 What should be kept the same when comparing two conditions?

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-204: Use an appropriate tool to collect a repeatable measurement.		/3
AQ-205: Show one quality or safety check before recording the result.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-206 Collect at least three values and calculate a useful summary such as mean, range, rate or percentage.

AQ-207 Graph or tabulate the result and identify the main pattern.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-208 Identify one limitation in your investigation and explain how it could affect the result.

AQ-209 Recommend one management or design change based on the evidence.

AQ-210 Design a follow-up test that would strengthen the conclusion.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-211 Define or explain one core idea related to shapes, sides and garden spaces.

AQ-212 Why is a fair comparison important when studying the garden bed or path?

AQ-213 What should be kept the same when comparing two conditions?

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-214: Use an appropriate tool to collect a repeatable shape or length.		/3
AQ-215: Show one quality or safety check before recording the result.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-216 Collect at least three values and calculate a useful summary such as mean, range, rate or percentage.

AQ-217 Graph or tabulate the result and identify the main pattern.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-218 Identify one limitation in your investigation and explain how it could affect the result.

AQ-219 Recommend one management or design change based on the evidence.

AQ-220 Design a follow-up test that would strengthen the conclusion.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-221 Define or explain one core idea related to treatment-control comparisons and fair testing.

AQ-222 Why is a fair comparison important when studying the grid cell?

AQ-223 What should be kept the same when comparing two conditions?

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-224: Use an appropriate tool to collect a repeatable treatment response.		/3
AQ-225: Show one quality or safety check before recording the result.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-226 Collect at least three values and calculate a useful summary such as mean, range, rate or percentage.

AQ-227 Graph or tabulate the result and identify the main pattern.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-228 Identify one limitation in your investigation and explain how it could affect the result.

AQ-229 Recommend one management or design change based on the evidence.

AQ-230 Design a follow-up test that would strengthen the conclusion.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-231 Define or explain one core idea related to access, moisture and resource flow in keyhole beds.

AQ-232 Why is a fair comparison important when studying the keyhole bed?

AQ-233 What should be kept the same when comparing two conditions?

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-234: Use an appropriate tool to collect a repeatable system observation.		/3
AQ-235: Show one quality or safety check before recording the result.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-236 Collect at least three values and calculate a useful summary such as mean, range, rate or percentage.

AQ-237 Graph or tabulate the result and identify the main pattern.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-238 Identify one limitation in your investigation and explain how it could affect the result.

AQ-239 Recommend one management or design change based on the evidence.

AQ-240 Design a follow-up test that would strengthen the conclusion.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-241 Define or explain one core idea related to temperature, moisture, volume and maturity.

AQ-242 Why is a fair comparison important when studying the compost pile?

AQ-243 What should be kept the same when comparing two conditions?

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-244: Use an appropriate tool to collect a repeatable compost indicator.		/3
AQ-245: Show one quality or safety check before recording the result.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-246 Collect at least three values and calculate a useful summary such as mean, range, rate or percentage.

AQ-247 Graph or tabulate the result and identify the main pattern.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-248 Identify one limitation in your investigation and explain how it could affect the result.

AQ-249 Recommend one management or design change based on the evidence.

AQ-250 Design a follow-up test that would strengthen the conclusion.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-251 Define or explain one core idea related to flow rate, uniformity and applied water.

AQ-252 Why is a fair comparison important when studying the drip emitter?

AQ-253 What should be kept the same when comparing two conditions?

Gravity Drip Garden - PRACTICAL & DATA

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-254: Use an appropriate tool to collect a repeatable flow rate.		/3
AQ-255: Show one quality or safety check before recording the result.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-256 Collect at least three values and calculate a useful summary such as mean, range, rate or percentage.

AQ-257 Graph or tabulate the result and identify the main pattern.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-258 Identify one limitation in your investigation and explain how it could affect the result.

AQ-259 Recommend one management or design change based on the evidence.

AQ-260 Design a follow-up test that would strengthen the conclusion.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-261 Define or explain one core idea related to performance across vertical levels.

AQ-262 Why is a fair comparison important when studying the vertical planter?

AQ-263 What should be kept the same when comparing two conditions?

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-264: Use an appropriate tool to collect a repeatable level response.		/3
AQ-265: Show one quality or safety check before recording the result.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-266 Collect at least three values and calculate a useful summary such as mean, range, rate or percentage.

AQ-267 Graph or tabulate the result and identify the main pattern.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-268 Identify one limitation in your investigation and explain how it could affect the result.

AQ-269 Recommend one management or design change based on the evidence.

AQ-270 Design a follow-up test that would strengthen the conclusion.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-271 Define or explain one core idea related to crop-family sequence and soil cover.

AQ-272 Why is a fair comparison important when studying the rotation record?

AQ-273 What should be kept the same when comparing two conditions?

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-274: Use an appropriate tool to collect a repeatable rotation interval.		/3
AQ-275: Show one quality or safety check before recording the result.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-276 Collect at least three values and calculate a useful summary such as mean, range, rate or percentage.

AQ-277 Graph or tabulate the result and identify the main pattern.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-278 Identify one limitation in your investigation and explain how it could affect the result.

AQ-279 Recommend one management or design change based on the evidence.

AQ-280 Design a follow-up test that would strengthen the conclusion.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-281 Define or explain one core idea related to infiltration, irrigation and water productivity.

AQ-282 Why is a fair comparison important when studying the water-smart bed?

AQ-283 What should be kept the same when comparing two conditions?

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-284: Use an appropriate tool to collect a repeatable water metric.		/3
AQ-285: Show one quality or safety check before recording the result.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-286 Collect at least three values and calculate a useful summary such as mean, range, rate or percentage.

AQ-287 Graph or tabulate the result and identify the main pattern.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-288 Identify one limitation in your investigation and explain how it could affect the result.

AQ-289 Recommend one management or design change based on the evidence.

AQ-290 Design a follow-up test that would strengthen the conclusion.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-291 Define or explain one core idea related to flow, pH, EC and root-zone condition.

AQ-292 Why is a fair comparison important when studying the soilless system?

AQ-293 What should be kept the same when comparing two conditions?

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-294: Use an appropriate tool to collect a repeatable system reading.		/3
AQ-295: Show one quality or safety check before recording the result.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-296 Collect at least three values and calculate a useful summary such as mean, range, rate or percentage.

AQ-297 Graph or tabulate the result and identify the main pattern.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-298 Identify one limitation in your investigation and explain how it could affect the result.

AQ-299 Recommend one management or design change based on the evidence.

AQ-300 Design a follow-up test that would strengthen the conclusion.

Student reflection: What evidence from this assessment are you most confident about?

MY GROWMEAL(TM) ASSESSMENT SUMMARY

My strongest area: _____

One skill I will improve: _____

One mission I want to try again: _____

Teacher feedback:
